

Matrix Embedding with LSBM and Hamming Codes

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1. True Embedding with LSBM and Hamming Codes

Approach

- Implemented matrix embedding explicitly by generating parity-check matrices H for Hamming codes.
- **Syndrome Calculation:** Compute $s = H \cdot y \pmod{2}$. The required change is $m - s \pmod{2}$.
- **Modification:** Find the index to change based on the difference.
- **LSBM:** In LSB Matching, changing the LSB equates to randomly adding or subtracting 1 from the pixel intensity, ensuring boundaries (like 0 and x_{max}) are respected.

Advantages

- Significantly increases embedding efficiency e compared to baseline steganography.
- Reduces the number of noticeable modifications (changes) in the cover object.

2. Simulating Hamming Codes

Simulation Strategy

- Each Hamming code block changes with probability $P(\text{change}) = \frac{2^k - 1}{2^k}$.
- The total number of modified blocks can be sampled directly using a Binomial distribution:
 $\sim \text{Binomial}\left(N_{\text{blocks}}, \frac{2^k - 1}{2^k}\right)$.
- Solves the need to explicitly execute code matrices over covers, massively increasing simulation speed.

Interpolation for Target α

- To achieve non-integer embedding rates α , we linearly interpolate the change probabilities per element of the two neighboring Hamming codes k and $k + 1$.

3. Performance and Precision Evaluation

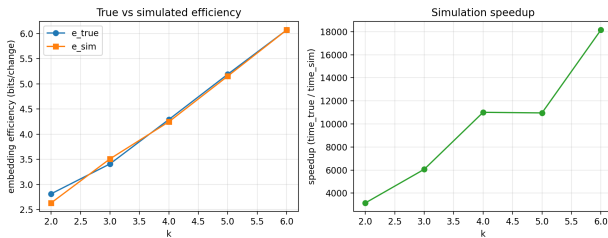
Speed and Precision (Simulation vs. True Embedding)

- **Precision:** The simulated embedding efficiency e_{sim} matches the e_{true} of fully processed image blocks up to high statistical precision.
- **Speedup:** The simulation reduces evaluation times from the millisecond/second range down to magnitudes of microseconds ($> 10^3 \times$ speedup).

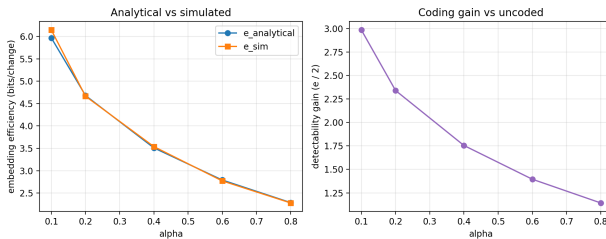
Detectability vs. Uncoded LSBM

- **Uncoded LSBM Baseline:** Random message flipping results in an embedding efficiency of $e \approx 2$ (bits embedded per change).
- **Coded Integrity:** The calculated analytical e progressively climbs well above 2, scaling directly with block sizes (reducing α).
- **Detectability Gain:** Because detectability correlates directly with the number of changes, our steganography achieves a significant drop in detectability (often reducing changes by multiples).

4. Benchmark for different k



5. Benchmark for different α



Summary

- **Completed Implementation:** Fully functional LSB Matching based upon matrix embedding with Hamming codes.
- **Fast Simulator derived:** Allows generating theoretical impacts for arbitrary embedding rates α via local interpolation.
- **Results:** We empirically validated the efficiency of Hamming codes, proving they are substantially safer from detectability and evaluating precisely correctly compared to direct code simulations.